Quantification: 872_wt_hp_z06
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 11 |
| interior lacuna count | 10 |
| frame-edge lacuna count | 1 |

Interior lacunae only: mean, median and sample SD (n = 10)
----------------------------------------------------------

| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 1562.4 | 1410.5 | 777.755 | px^2 |
| major_axis_length_px | lacuna shape | 75.3604 | 74.0553 | 23.8177 | px |
| minor_axis_length_px | lacuna shape | 26.8961 | 23.3362 | 8.1892 | px |
| aspect_ratio | lacuna shape | 2.889 | 2.8577 | 0.8048 | unitless |
| eccentricity | lacuna shape | 0.9134 | 0.9355 | 0.0793 | unitless |
| solidity | lacuna shape | 0.8681 | 0.8856 | 0.0523 | unitless |
| roots_count | headline | 10.9 | 11 | 4.9989 | count |
| ring_length_r30_px | headline | 500.7 | 472.5 | 116.563 | px |
| ring_length_r60_px | reported | 1251.6 | 1167.5 | 300.043 | px |
| owned_length_px | ownership-dependent | 3320.17 | 3155.23 | 677.639 | px |
| edge_count | ownership-dependent | 207.5 | 204 | 51.4485 | count |
| mean_edge_length_px | ownership-dependent | 16.1887 | 16.0665 | 1.248 | px |
| perimeter_px | normalised | 199.93 | 204.217 | 62.4899 | px |
| ring_area_r30_px2 | normalised | 8113 | 8476 | 1378.18 | px^2 |
| ring_area_r60_px2 | normalised | 20923.1 | 22211 | 2899.92 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9893 | 1 | 0.0338 | unitless |
| in_frame_fraction_r60 | normalised | 0.981 | 1 | 0.0602 | unitless |
| ring_density_r30 | normalised | 0.0623605 | 0.0660765 | 0.0123922 | px^-1 |
| ring_density_r60 | normalised | 0.060771 | 0.0617304 | 0.0152229 | px^-1 |
| roots_per_100px_perimeter | normalised | 5.3918 | 5.8302 | 1.8186 | per 100 px |
| ring_attached_length_r30_px | network v2 | 469.6 | 446 | 116.156 | px |
| ring_attached_length_r60_px | network v2 | 1099.1 | 871 | 375.137 | px |
| ring_length_w_r30_px | network v2 | 569.488 | 538.318 | 135.423 | px |
| ring_length_w_r60_px | network v2 | 1426.55 | 1324.29 | 344.212 | px |
| sholl_crossings_r10 | network v2 | 11.1 | 11 | 3.1073 | count |
| sholl_crossings_r20 | network v2 | 14.1 | 12.5 | 3.8137 | count |
| sholl_crossings_r30 | network v2 | 15.8 | 16 | 3.9944 | count |
| loop_count | loops, report only | 29.5 | 23.5 | 13.4268 | count |
| loop_edge_share | loops, report only | 0.5188 | 0.4845 | 0.1322 | share |
| ring_width_mean_r30_px | width | 7.2659 | 7.1719 | 0.5794 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 780579 | px^2 |
| total_skeleton_length_px | 38600 | px |
| canalicular_length_density_per_px | 0.0494505 | px^-1 |
| junction_count | 1289 | count |
| junction_density_per_px2 | 0.00165134 | px^-2 |
| lacuna_count | 11 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.409e-05 | px^-2 |
| skeleton_component_count | 149 | count |
| mean_component_length_px | 259.06 | px |
| median_component_length_px | 18 | px |
| field_length_density_w_per_px | 0.0561396 | px^-1 |
| field_density_without_flagged_per_px | 0.0384861 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 10 | px |
| unattached_component_count | 1 | count |
| canalicular_length_density_attached_per_px | 0.0494505 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0494633 | px^-1 |
| excluded_area_px2 | 251704 | px^2 |
| analysed_field_area_px2 | 796872 | px^2 |
| loop_count | 410 | count |
| loop_edge_share | 0.5534 | share |
| edge_count | 2732 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.6024 | px |
| width_median_px | 8 | px |
| width_p10_px | 4 | px |
| width_p90_px | 10.7703 | px |
| width_pixels_used | 39416 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2178 | 97.2133 | 31.5882 | 3.0775 | 0.9457 | 0.7666 | -1.4507 | 132.953 | 694.614 | False |
| 2 | 1532 | 90.3989 | 22.6854 | 3.9849 | 0.968 | 0.8855 | -1.4786 | 133.692 | 263.431 | False |
| 3 | 1458 | 82.9994 | 23.987 | 3.4602 | 0.9573 | 0.8423 | -1.4301 | 272.217 | 418.857 | False |
| 4 | 2267 | 64.9571 | 46.3919 | 1.4002 | 0.6999 | 0.896 | -1.1815 | 380.946 | 282.045 | False |
| 5 | 3165 | 121.397 | 33.5624 | 3.6171 | 0.961 | 0.8709 | -1.4322 | 386.945 | 67.1896 | False |
| 6 | 743 | 47.9512 | 20.7766 | 2.3079 | 0.9013 | 0.8888 | -1.5064 | 382.218 | 839.734 | False |
| 7 | 712 | 46.6981 | 20.2555 | 2.3054 | 0.901 | 0.8856 | -1.5028 | 499.643 | 510.035 | False |
| 8 | 1363 | 80.3055 | 21.9882 | 3.6522 | 0.9618 | 0.8961 | -1.3459 | 561.869 | 697.237 | False |
| 9 | 669 | 39.9711 | 24.8769 | 1.6068 | 0.7827 | 0.7707 | -1.1357 | 585.831 | 14.0269 | True |
| 10 | 1360 | 67.805 | 25.7054 | 2.6378 | 0.9254 | 0.9484 | -1.4973 | 614.482 | 323.551 | False |
| 11 | 846 | 53.8777 | 22.02 | 2.4468 | 0.9127 | 0.8004 | 1.5387 | 642.791 | 211.617 | False |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 10 | 465 | 967 | 3209.31 | 211 | 15.21 | 7.8301 |
| 2 | 12 | 355 | 826 | 2856.26 | 176 | 16.2288 | 6.7613 |
| 3 | 18 | 571 | 1575 | 3785.2 | 238 | 15.9042 | 7.6043 |
| 4 | 14 | 611 | 1560 | 3024.19 | 197 | 15.3512 | 8.193 |
| 5 | 18 | 743 | 1764 | 4428.6 | 298 | 14.8611 | 7.3047 |
| 6 | 7 | 425 | 1107 | 3971.9 | 223 | 17.8112 | 6.9899 |

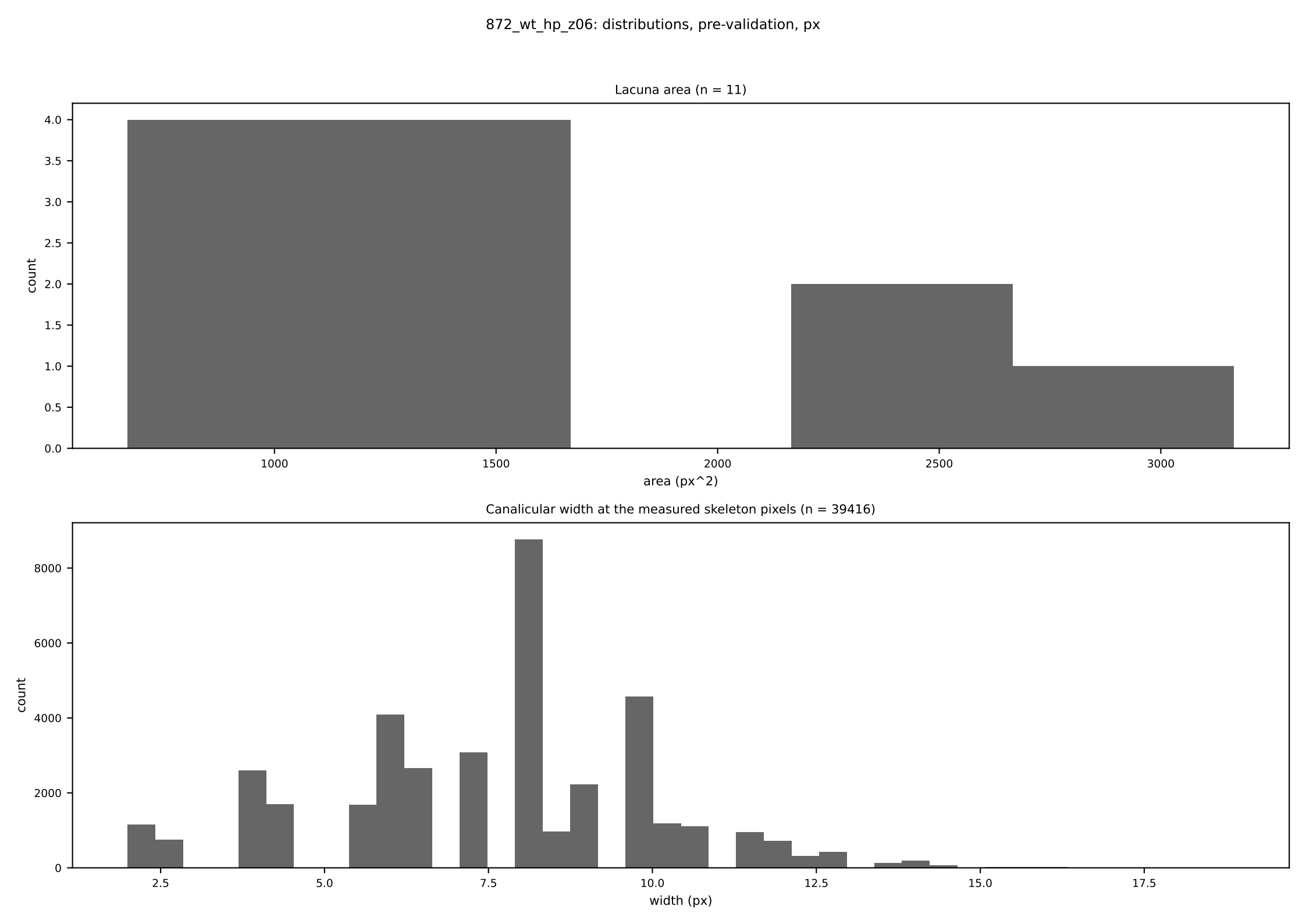
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 7 | 4 | 406 | 1128 | 2127.61 | 124 | 17.1582 | 6.5184 |
| 8 | 12 | 480 | 1207 | 2860.76 | 159 | 17.9922 | 7.0391 |
| 9 | 8 | 353 | 919 | 2462.03 | 161 | 15.2921 | 7.9105 |
| 10 | 4 | 414 | 1049 | 3101.16 | 183 | 16.9462 | 6.5847 |
| 11 | 10 | 537 | 1333 | 3836.7 | 266 | 14.4237 | 7.8334 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 292.451 | 10015 | 25442 | 1 | 1 | 0.0464304 | 0.038008 | 3.4194 |
| 2 | 218.853 | 8881 | 23379 | 1 | 1 | 0.039973 | 0.0353308 | 5.4831 |
| 3 | 215.095 | 8592 | 22238 | 1 | 1 | 0.0664572 | 0.0708247 | 8.3684 |
| 4 | 212.409 | 8564 | 22184 | 1 | 1 | 0.0713452 | 0.0703209 | 6.5911 |
| 5 | 301.179 | 9919 | 22490 | 0.893 | 0.8096 | 0.0749067 | 0.0784349 | 5.9765 |
| 6 | 123.154 | 6283 | 18209 | 1 | 1 | 0.0676429 | 0.0607941 | 5.6839 |
| 7 | 119.983 | 6180 | 18000 | 1 | 1 | 0.0656958 | 0.0626667 | 3.3338 |
| 8 | 196.024 | 8388 | 22420 | 1 | 1 | 0.0572246 | 0.0538359 | 6.1217 |
| 9 | 129.29 | 4152 | 11045 | 0.6578 | 0.6072 | 0.0850193 | 0.0832051 | 6.1876 |
| 10 | 167.539 | 7464 | 18130 | 1 | 1 | 0.0554662 | 0.0578599 | 2.3875 |
| 11 | 152.61 | 6844 | 16739 | 1 | 1 | 0.0784629 | 0.0796344 | 6.5526 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 456 | 902 | 528.262 | 1086.45 | 10 | 12 | 10 |
| 2 | 353 | 732 | 409.706 | 944.625 | 7 | 12 | 11 |
| 3 | 565 | 1554 | 641.789 | 1784.43 | 14 | 18 | 16 |
| 4 | 562 | 1456 | 701.357 | 1795.73 | 13 | 16 | 21 |
| 5 | 704 | 1737 | 852.34 | 2010.25 | 17 | 22 | 21 |
| 6 | 388 | 803 | 482.262 | 1285 | 11 | 10 | 16 |
| 7 | 333 | 834 | 452.149 | 1284.97 | 7 | 11 | 15 |
| 8 | 436 | 817 | 548.375 | 1363.57 | 11 | 16 | 20 |
| 9 | 347 | 887 | 409.806 | 1051.35 | 6 | 11 | 10 |
| 10 | 387 | 840 | 463.291 | 1183.31 | 9 | 11 | 12 |
| 11 | 512 | 1316 | 615.345 | 1527.2 | 12 | 13 | 16 |


872_wt_hp_z06: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 39416)
count
width (px)